

THE MOMENTS OF A FUNCTION OF TRACES OF A MATRIX WITH A MULTIVARIATE SYMMETRIC NORMAL DISTRIBUTION

Takesi Hayakawa* and Yumiko Kikuchi

Institute of Statistical Mathematics

Japan

Key Words: Moment, multivariate symmetric normal distribution, Wishart distribution, zonal polynomial.

SUMMARY

The moments of a function of traces of a matrix with a multivariate symmetric normal distribution are derived by the use of zonal polynomials of a matrix argument and these are tabulated up to 8 degrees.

1. Introduction

Let Z be a $p \times p$ symmetric matrix with a probability density function

$$(1) \quad f(Z) = \exp \left\{ -\frac{1}{2} \text{tr } Z^2 \right\} / 2^{p/2} \pi^{p(p+1)/4},$$

then we say that Z has a multivariate symmetric normal distribution. This matrix Z plays an important role in the derivation of the asymptotic expansion of the distribution of a statistic which depends on a Wishart matrix (Nagao (1973) etc), and Olson and Uppuluri (1970) showed that the multivariate symmetric normal distribution is that of a symmetric random matrix which is invariant over the orthogonal transformation. In the derivation process of the asymptotic expansion of the distribution we need several moments of functions of Z . It is not difficult to calculate the moments of low degrees, but it becomes quite laborious to handle these for higher degrees. Because these moments are so important, we tabulate them up to 8 degrees in Table II.

Concerning this moment problems, Nel, D.G. (1977) and Nel, H.M. (1977) have tabulated the expectation of all possible combinations of powers of the elements, as well as the elementary symmetric functions of these matrices, up to 4th degrees for the general multivariate symmetric normal distribution, of which (1) is a special case.

2. Calculation of moments

Let S be a $p \times p$ positive definite symmetric matrix distributed with a Wishart dis-

*Presently at Hitotsubashi University

tribution $W(\Sigma, n)$. To calculate the moments we cite the tables of zonal polynomials of matrix argument S which have been tabulated up to 12 by Parkhurst and James (1974).

It is well-known (Constantine (1963)) that the expectation of a zonal polynomial $C_\kappa(S)$ corresponding to a partition κ of k into not more than p parts $\kappa = (k_1, k_2, \dots, k_p)$, $k_1 \geq k_2 \geq \dots \geq k_p \geq 0$, with respect to a Wishart distribution $W(\Sigma, n)$, is expressed as follows.

$$(2) \quad E_S [C_\kappa(S)] = 2^k \left(\frac{n}{2}\right)_\kappa C_\kappa(\Sigma),$$

where

$$\left(\frac{n}{2}\right)_\kappa = \prod_{i=1}^p \left(\frac{n}{2} - \frac{i-1}{2}\right)_{k_i}, (a)_\ell = a(a+1) \dots (a+\ell-1).$$

Putting

$$(3) \quad Z_\kappa(S) = \frac{N}{\chi_{[2\kappa]}(1)} C_\kappa(S),$$

$$N = (2k)! / 2^k k!$$

$$\chi_{[2\kappa]}(1) = \frac{\prod_{i=1}^m (2k_i - 2k_j - i + j)}{\prod_{i=1}^m (2k_i + m - i)!}$$

and

$$z'_s = (Z_{(k)}(S), Z_{(k-1,1)}(S), \dots, Z_{(1^k)}(S)),$$

$$c'_s = (C_{(k)}(S), C_{(k-1,1)}(S), \dots, C_{(1^k)}(S)),$$

$$z'_\Sigma = (Z_{(k)}(\Sigma), Z_{(k-1,1)}(\Sigma), \dots, Z_{(1^k)}(\Sigma)),$$

$$(4) \quad c'_\Sigma = (C_{(k)}(\Sigma), C_{(k-1,1)}(\Sigma), \dots, C_{(1^k)}(\Sigma)),$$

$$D_d = \text{diag} (N/\chi_{[2(k)]}(1), N/\chi_{[2(k-1,1)]}(1), \dots, N/\chi_{[2(1^k)]}(1)),$$

$$M_d = \text{diag} (2^k \left(\frac{n}{2}\right)_{(k)}, 2^k \left(\frac{n}{2}\right)_{(k-1,1)}, \dots, 2^k \left(\frac{n}{2}\right)_{(1^k)}),$$

$$b'_s = (s_1^k, s_1^{k-2} s_2, \dots, s_k)$$

$$b'_\Sigma = (\sigma_1^k, \sigma_1^{k-2} \sigma_2, \dots, \sigma_k)$$

where $s_j = \text{tr} S^j$, and $\sigma_j = \text{tr} \Sigma^j$, we can express z_s in terms of b_s as

$$(5) \quad z_s = A_k b_s, \quad b_s = A_k^{-1} z_s.$$

Parkhurst and James tabulated these A_k 's for $k = 1(1)12$. As the coefficients of A_k satisfy orthogonality relations, James (1964),

$$(6) \quad \sum_{\nu} z_{k\nu} z_{\lambda\nu} / z_{(k)\nu} = \delta_{k\lambda} N / \chi[2(k)] \quad (1),$$

by putting

$$(7) \quad B = (b_{k\nu}), \quad b_{k\nu} = z_{k\nu} \chi[2(k)]^{(1)/N} z_{(k)\nu},$$

where $z_{(k)\nu}$ is the coefficient of top zonal polynomial $Z_{(k)}(S)$, we have

$$A_k B' = I_m, \text{ that is, } B' = A_k^{-1}.$$

For $\lambda = (k)$ in (6), we have

$$(8) \quad \sum_{\nu} z_{k\nu} = \delta_{k(k)} N / \chi[2k] \quad (1).$$

Using these notations, (2) is expressed as

$$\begin{aligned} (9) \quad E[b_s] &= A_k^{-1} E[z_s] = A_k^{-1} D_d E[c_s] \\ &= A_k^{-1} D_d M_d c_{\Sigma} = A_k^{-1} D_d M_d D_d^{-1} z_{\Sigma} \\ &= A_k^{-1} M_d A_k b_{\Sigma}. \end{aligned}$$

Since M_d is a diagonal matrix whose diagonal elements are functions of n , we expand M_d as

$$(10) \quad M_d = n^{k_1} + n^{k-1} B_{k-1}^{(k)} + \dots + n B_1^{(k)}.$$

Therefore $A_k^{-1} M_d A_k$ is expressed as

$$(11) \quad A_k^{-1} M_d A_k = n^{k_1} + n^{k-1} A_{k-1}^{(k)} + \dots + n A_1^{(k)},$$

where $A^{(k)} = A_k^{-1} B_\ell^{(k)} A_k$, $\ell = 1, 2, \dots, k-1$.

Fujikoshi (1973) tabulated $A^{(k)}$'s for $k = 1(1)6$.

It is of interest to note that (12) holds for any κ . Summing the row elements $c_{\kappa\nu}$ of $C = A_k^{-1} M_d A_k$, we obtain

$$\begin{aligned}
 (12) \quad \sum_{\mu} c_{\kappa\mu} &= \sum_{\mu} \left\{ \sum_{\nu} b_{\nu\kappa} m_{\nu} z_{\nu\mu} \right\} \\
 &= \sum_{\nu} b_{\nu\kappa} m_{\nu} \sum_{\mu} z_{\nu\mu} \\
 &= \sum_{\nu} b_{\nu\kappa} m_{\nu} \delta_{\nu(k)} N/\chi[2\nu] (1) \\
 &= b_{(k)\kappa} m_{(k)} N/\chi[2(k)] (1) = m_{(k)} = 2^k \left(\frac{n}{2}\right)_k \\
 &= n(n+2)(n+4) \dots (n+2k-2).
 \end{aligned}$$

This implies that the sum of the elements of any row of $A^{(k)}$ should be equal to the coefficient of n^k of the right hand side of (12), and we can use this to check the accuracy of the matrix calculation.

Putting $\Sigma = I_p$ in b_{Σ} and rearranging the elements of b_I in the order of the power of p , that is, $p' = (p^k, p^{k-1}, \dots, p)$, we can define a new matrix $G^{(k)}$ by (9),

$$(13) \quad B(b_s) = G^{(k)} p.$$

Define $n' = (n^k, n^{k-1}, \dots, n)$ and corresponding to each element $E(s_1^{\nu_1} s_2^{\nu_2} \dots s_k^{\nu_k})$, $\nu = (\nu_1, \dots, \nu_k)$, $1 \cdot \nu_1 + 2 \cdot \nu_2 + \dots + k \cdot \nu_k = k$, of $E(b_s)$ we define a new matrix $H_{\nu}^{(k)}$ as follows:

$$(14) \quad E(b_s) = (n' H_{(k)}^{(k)} p, n' H_{(k-2,1)}^{(k)} p, \dots, n' H_{(0^{k-1},1)}^{(k)} p)'.$$

All these matrices $H_{\nu}^{(k)}$, $k = 1(1)12$, become symmetric matrices. For example of $k = 4$,

$$E(s_1^4) = n' H_{(4)}^{(4)} p = n' \begin{bmatrix} 1 & & & \\ & 12 & & \\ & & 44 & \\ & & & 48 \end{bmatrix} p$$

$$= p^4 n^4 + 12p^3 n^3 + 44p^2 n^2 + 48pn,$$

$$E\left(\begin{smallmatrix} 2 \\ 1 & 2 \end{smallmatrix}\right) = n' H_{(21)}^{(4)} p = n' \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 10 & 0 \\ 0 & 10 & 10 & 24 \\ 0 & 0 & 24 & 24 \end{bmatrix} p$$

$$= p^3 n^4 + (p^4 + p^3 + 10p^2)n^3 + (10p^3 + 10p^2 + 24p)n^2 \\ + (24p^2 + 24p)n$$

etc.

The diagonal elements of $H_{(4)}^{(4)}$ are the coefficients of the polynomial

$$n(n+2)(n+4)(n+6),$$

and the sums of elements of any row of each matrix are the same for all matrices. This fact can be used as a check.

The $H_{\nu}^{(k)}$'s are tabulated up to $k = 6$ in Table 1.

To calculate the moments of a function of Z , which has a probability density function $f(Z) = \exp. \{-\text{tr } Z^2 / 2\} / 2^{p/2} \pi^{p(p+1)/4}$, we consider a transformation

$$(15) \quad Z_n = (S - nI_p) / \sqrt{2n}.$$

The limiting probability density function of Z_n is $f(Z)$.

We use the following lemma to handle the moments of a limiting distribution

Lemma (Rao, (1968)).

Let α_m be the r -th moment of a distribution F_n and α_m converges to α_r , where α_m and α_r are finite for all r and large n . Then, if the distribution F_n converges to a distribution F , α_r is the r -th moment of F .

As an example, we show how to calculate the moment $E[\text{tr } Z^4 \text{tr } Z^2]$.

$$\text{tr } Z_n^4 \text{tr } Z_n^2 = \frac{1}{8n^3} [s_4 s_2 - n(4s_3 s_2 + 2s_4 s_1) + n^2(6s_2^2 + 8s_3 s_1 + ps_4) \\ - n^3(16s_2 s_1 + 4ps_3) + n^4(7ps_2 + 8s_1^2)]$$

$$-n^5(6ps_1) + n^6 p^2],$$

where $s_j = \text{tr} S^j$. Then the expectation $E[\text{tr} Z_n^4 \text{tr} Z_n^2]$ is expressed by the use of the tables of the $H_{\nu}^{(k)}$'s for $k = 1(1)6$ as

$$\begin{aligned} E[\text{tr} Z_n^4 \text{tr} Z_n^2] &= \frac{1}{8}(2p^5 + 7p^4 + 26p^3 + 45p^2 + 40p) \\ &\quad + \frac{1}{8n}(p^6 + 7p^5 + 99p^4 + 477p^3 + 884p^2 + 712p) \\ &\quad + \frac{1}{8n^2}(16p^5 + 136p^4 + 736p^3 + 1608p^2 + 1344p). \end{aligned}$$

Since the limiting distribution of Z_n is that of Z , we have the moment

$$\begin{aligned} E[\text{tr} Z^4 \text{tr} Z^2] &= \lim_{n \rightarrow \infty} E[\text{tr} Z_n^4 \text{tr} Z_n^2] \\ &= \frac{1}{8}(2p^5 + 7p^4 + 26p^3 + 45p^2 + 40p). \end{aligned}$$

We calculated these up to 12 degrees and tabulated them up to 8 degrees in Table II.

I. Tables of $H^{(k)}$, $k = 1(1)6$

$k = 2$

$$H_{(2)} = \begin{bmatrix} 1 & & \\ & 2 & \end{bmatrix}$$

$$H_{(0 \ 1)} = \begin{bmatrix} & 1 \\ 1 & 1 \end{bmatrix}$$

$k = 3$

$$H_{(3)} = \begin{bmatrix} 1 & & & \\ & 6 & & \\ & & 8 & \end{bmatrix}$$

$$H_{(1 \ 1)} = \begin{bmatrix} & 1 & & \\ & 1 & 4 & \\ 1 & & 4 & 4 \end{bmatrix}$$

$$H_{(0^2 \ 1)} = \begin{bmatrix} & & 1 \\ & 3 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

$k = 4$

$$H_{(4)} = \begin{bmatrix} 1 & & & & \\ & 12 & & & \\ & & 44 & & \\ & & & 48 & \end{bmatrix}$$

$$H_{(2 \ 1)} = \begin{bmatrix} & & 1 & & \\ & 1 & 10 & & \\ & 10 & 10 & 24 & \\ 1 & & 24 & 24 \end{bmatrix}$$

I. Tables of $H^{(k)}$, $k = 1(1)6$ (Continued)

$$H_{(0\ 2)} = \begin{bmatrix} & & 1 & \\ & 2 & 2 & 8 \\ 1 & 2 & 21 & 20 \\ & 8 & 20 & 20 \end{bmatrix}$$

$$H_{(1\ 0\ 1)} = \begin{bmatrix} & & & 1 & \\ & 3 & 3 & 6 & \\ & 3 & 22 & 18 & \\ 1 & 6 & 18 & 24 & \end{bmatrix}$$

$$H_{(0^3\ 1)} = \begin{bmatrix} & & & & 1 \\ & & 6 & 6 & \\ & 6 & 17 & 21 & \\ 1 & 6 & 21 & 20 & \end{bmatrix}$$

 $k = 5$

$$H_{(5)} = \begin{bmatrix} 1 & & & & & \\ & 20 & & & & \\ & & 140 & & & \\ & & & 400 & & \\ & & & & 384 & \end{bmatrix}$$

$$H_{(3\ 1)} = \begin{bmatrix} & & & 1 & & \\ & & & 1 & 18 & \\ & 18 & & 18 & 104 & \\ & & 104 & 104 & 192 & \\ & & & 192 & 192 & \end{bmatrix}$$

$$H_{(1\ 2)} = \begin{bmatrix} & & & 1 & & \\ & 2 & 2 & 16 & & \\ 1 & 2 & 37 & 36 & 64 & \\ & 16 & 36 & 188 & 160 & \\ & & 64 & 160 & 160 & \end{bmatrix}$$

$$H_{(2\ 0\ 1)} = \begin{bmatrix} & & & & 1 & \\ & 3 & 3 & 14 & & \\ & 3 & 46 & 42 & 48 & \\ 1 & 14 & 42 & 200 & 144 & \\ & & 48 & 144 & 192 & \end{bmatrix}$$

$$H_{(0\ 1\ 1)} = \begin{bmatrix} & & & & 1 & \\ & & 4 & 4 & 12 & \\ & 4 & 9 & 67 & 60 & \\ 1 & 4 & 67 & 160 & 168 & \\ & 12 & 60 & 168 & 144 & \end{bmatrix}$$

$$H_{(10^2\ 1)} = \begin{bmatrix} & & & & & 1 & \\ & & & 6 & 6 & 8 & \\ & 6 & 17 & 69 & 48 & & \\ & 6 & 69 & 156 & 168 & & \\ 1 & 8 & 48 & 168 & 160 & & \end{bmatrix}$$

$$H_{(0^4\ 1)} = \begin{bmatrix} & & & & & & 1 \\ & & & & 10 & 10 & \\ & & 20 & 55 & 65 & & \\ & 10 & 55 & 175 & 160 & & \\ 1 & 10 & 65 & 160 & 148 & & \end{bmatrix}$$

I. Tables of $H^{(k)}$, $k = 1(1)6$ (Continued) $k = 6$

$$H_{(6)} = \begin{bmatrix} 1 & & & & & \\ & 30 & & & & \\ & & 340 & & & \\ & & & 1800 & & \\ & & & & 4384 & \\ & & & & & 3840 \end{bmatrix}$$

$$H_{(4\ 1)} = \begin{bmatrix} & & 1 & & & \\ 1 & & & & & \\ & 1 & 28 & & & \\ & 28 & 28 & 284 & & \\ & & 284 & 284 & 1232 & \\ & & & 1232 & 1232 & 1920 \\ & & & & 1920 & 1920 \end{bmatrix}$$

$$H_{(2\ 2)} = \begin{bmatrix} & & & 1 & & \\ & 2 & 2 & 26 & & \\ 1 & 2 & 57 & 56 & 224 & \\ & 26 & 56 & 558 & 520 & 640 \\ & & 224 & 520 & 2040 & 1600 \\ & & & 640 & 1600 & 1600 \end{bmatrix}$$

$$H_{(3\ 0\ 1)} = \begin{bmatrix} & & & & 1 & \\ & 3 & 3 & 24 & & \\ 1 & 3 & 76 & 72 & 188 & \\ & 24 & 72 & 660 & 564 & 480 \\ & & 188 & 564 & 2192 & 1440 \\ & & & 480 & 1440 & 1920 \end{bmatrix}$$

$$H_{(0\ 3)} = \begin{bmatrix} & & & & & 1 \\ & & 3 & 3 & 24 & \\ & 3 & 6 & 87 & 84 & 160 \\ 1 & 3 & 87 & 181 & 824 & 704 \\ & 24 & 84 & 824 & 1788 & 1664 \\ & & 160 & 704 & 1664 & 1312 \end{bmatrix}$$

I. Tables of $H^{(k)}$, $k = 1(1)6$ (Continued)

$$H_{(1\ 1\ 1)} = \begin{bmatrix} & & & 1 & & \\ & & 4 & 4 & 22 & \\ & 4 & 9 & 107 & 100 & 120 \\ & 4 & 107 & 250 & 838 & 600 \\ 22 & 100 & 838 & 1744 & 1680 & \\ & 120 & 600 & 1680 & 1440 & \end{bmatrix}$$

$$H_{(2\ 0^2\ 1)} = \begin{bmatrix} & & & 1 & & \\ & & 6 & 6 & 18 & \\ & 6 & 17 & 129 & 108 & 80 \\ & 6 & 129 & 326 & 858 & 480 \\ 18 & 108 & 858 & 1720 & 1680 & \\ & 80 & 480 & 1680 & 1600 & \end{bmatrix}$$

$$H_{(0^2\ 2)} = \begin{bmatrix} & & & & 1 & \\ & & & 6 & 6 & 18 \\ & & 11 & 24 & 161 & 144 \\ & 6 & 24 & 318 & 726 & 726 \\ 1 & 6 & 161 & 726 & 1906 & 1584 \\ & 18 & 144 & 726 & 1584 & 1368 \end{bmatrix}$$

$$H_{(0\ 1\ 0\ 1)} = \begin{bmatrix} & & & & 1 & \\ & & & 7 & 7 & 16 \\ & & 12 & 29 & 163 & 136 \\ & 7 & 29 & 315 & 713 & 736 \\ 1 & 7 & 163 & 713 & 1892 & 1608 \\ & 16 & 136 & 736 & 1608 & 1344 \end{bmatrix}$$

$$H_{(1\ 0^3\ 1)} = \begin{bmatrix} & & & & 1 & \\ & & & 10 & 10 & 10 \\ & & 20 & 55 & 165 & 100 \\ & 10 & 55 & 375 & 710 & 650 \\ 1 & 10 & 165 & 710 & 1898 & 1600 \\ & 10 & 100 & 650 & 1600 & 1480 \end{bmatrix}$$

I. Tables of $H^{(k)}$, $k = 1(1)6$ (Continued)

$$H_{(0^5 1)} = \begin{bmatrix} & & & & & 1 \\ & & & & 15 & 15 \\ & & 50 & 135 & 155 & \\ & 50 & 262 & 787 & 701 & \\ 15 & 135 & 787 & 1827 & 1620 & \\ 1 & 15 & 155 & 701 & 1620 & 1348 \end{bmatrix}$$

II. Tables of expectations for Z

For example, $2 \ 1^2 \Rightarrow E[\text{tr } Z^2 (\text{tr } Z)^2] = \frac{1}{4}(2p^3 + 2p^2 + 8p)$.

$$\begin{bmatrix} 2 \\ 1^2 \end{bmatrix} \Rightarrow \frac{1}{2} \begin{bmatrix} p^2 & p \\ 1 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ 3 \ 1 \\ 2^2 \\ 2 \ 1^2 \\ 1^4 \end{bmatrix} \Rightarrow \frac{1}{4} \begin{bmatrix} p^4 & p^3 & p^2 & p \\ 0 & 2 & 5 & 5 \\ 0 & 0 & 6 & 6 \\ 1 & 2 & 5 & 4 \\ 0 & 2 & 2 & 8 \\ 0 & 0 & 12 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 6 \\ 5 \ 1 \\ 4 \ 2 \\ 4 \ 1^2 \\ 3^2 \\ 3 \ 2 \ 1 \\ 3 \ 1^3 \\ 2^3 \\ 2^2 \ 1^2 \\ 2 \ 1^4 \\ 1^6 \end{bmatrix} \Rightarrow \frac{1}{8} \begin{bmatrix} p^6 & p^5 & p^4 & p^3 & p^2 & p \\ 0 & 0 & 5 & 22 & 52 & 41 \\ 0 & 0 & 0 & 20 & 50 & 50 \\ 0 & 2 & 7 & 26 & 45 & 40 \\ 0 & 0 & 4 & 10 & 58 & 48 \\ 0 & 0 & 0 & 24 & 54 & 42 \\ 0 & 0 & 6 & 12 & 54 & 48 \\ 0 & 0 & 0 & 36 & 36 & 48 \\ 1 & 3 & 15 & 25 & 44 & 32 \\ 0 & 2 & 4 & 26 & 24 & 64 \\ 0 & 0 & 12 & 12 & 96 & 0 \\ 0 & 0 & 0 & 120 & 0 & 0 \end{bmatrix}$$

II. Tables of expectations for Z (Continued)

	p^8	p^7	p^6	p^5	p^4	p^3	p^2	p
8	0	0	0	14	93	374	690	509
7 1	0	0	0	0	70	308	728	574
6 2	0	0	5	27	134	357	665	492
6 1 ²	0	0	0	10	44	344	682	600
5 3	0	0	0	0	90	360	720	510
5 2 1	0	0	0	20	70	340	650	600
5 1 ³	0	0	0	0	120	300	780	480
4 ²	0	0	4	20	117	354	689	496
4 3 1	0	0	0	12	42	348	702	576
4 2 ²	0	2	9	57	155	397	580	480
4 2 1 ²	0	0	4	14	116	226	744	576
4 1 ⁴	0	0	0	24	60	636	576	384
3 ² 2	0	0	0	24	78	384	690	504
3 ² 1 ²	0	0	0	0	120	252	732	576
3 2 ² 1	0	0	6	18	138	246	696	576
3 2 1 ³	0	0	0	36	72	516	480	576
3 1 ⁵	0	0	0	0	360	360	960	0
2 ⁴	1	4	30	76	249	376	560	384
2 ³ 1 ²	0	2	6	54	98	400	352	768
2 ² 1 ⁴	0	0	12	24	252	240	1152	0
2 1 ⁶	0	0	0	120	120	1440	0	0
1 ⁸	0	0	0	0	1680	0	0	0

 $\Rightarrow \frac{1}{16}$

REFERENCES

- CONSTANTINE, A.G. (1963). "Some non-central distribution problems in multivariate analysis." *Ann. Math. Statist.* **34**, 1270-1285.
- FUJIKOSHI, Y. (1973). "Asymptotic formulas for the distributions of three statistics for multivariate linear hypothesis." *Ann. Inst. Statist. Math.*, **25**, 423-437.
- JAMES, A.T. (1964). "Distributions of matrix variates and latent roots derived from normal samples." *Ann. Math. Statist.*, **35**, 475-501.
- NAGAO, H. (1973). "On some test criteria for covariance matrix." *Ann. Statist.*, **1**, 700-709.
- NEL, D.G. (1977). "On the symmetric multivariate normal distribution and the asymptotic expansion of a Wishart matrix." *S. Afr. Statist. J.*, **12**, 145-159.
- NEL, H.M. (1977). "On distributions and moments associated with matrix normal distributions." Technical report No. 24, Dept. Mathematical Statistics, Univ. of the Orange Free State.
- OLSON, W.H. and UPPULURI, V.R. RAO (1970). "Characterization of the distribution of a random matrix by rotational invariance." *Sankhyā A*, **32**, 325-336.
- PARKHURST, A.M. and JAMES, A.T. (1974). "Zonal polynomials of order 1 through 12." Selected tables in Mathematical Statistics, Vol II, (H.L. Harter and D.B. Owens, Eds) American Mathematical Society, Providence.
- RAO, C.R. (1968). "*Linear statistical inference and its applications*." Wiley, New York.

Manuscript received 1978-03-28.